

Master's Programme in Bioinformatics

Welcome

Umeå University · 2026–2028

120 ECTS

2 Years

Research-led

Teaching

Bioinformatics

Careers

Department of Plant Physiology · Umeå University

Programme Leader: Nathaniel Street · nathaniel.street@umu.se

Your Programme Leader

Nathaniel Street



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Department of Plant Physiology



KBC Building, Umeå University



Research: Plant genomics, bioinformatics,
systems biology



plantgenie.org · upsc.se

Contact me for:



Questions about course content, assessments, or grades



Questions about the programme structure or progression



Concerns about your progress or wellbeing



Ideas, feedback, or suggestions for the programme



HPC/Kebnekaise technical issues → HPC2N support



University registration → Student Affairs Office



Accessibility needs → Umeå University Student Health

Office hours: posted on Canvas before each session

Why This Programme, Why Now

~10 petabytes

of genomic data generated annually and doubling every 18 months

#2 skill

'bioinformatics/computational biology' in life science job listings globally

200+ billion

DNA sequencing reads deposited in ENA/SRA — each one a question waiting to be answered

You are entering a field at an inflection point. The data exists. The questions are urgent. What is needed — what you are here to learn — is the capacity to work with it.

What We Are Trying to Achieve

1

Technical mastery

You will develop genuine, production-level bioinformatics skills — not a surface familiarity, but the ability to design, execute, and debug analyses on real data at scale.

2

Scientific judgement

You will develop the critical capacity to evaluate tools, methods, and results — including the limitations, assumptions, and potential failure modes of computational analyses.

3

Reproducible and open practice

You will work from day one in ways that are reproducible, documented, and FAIR-compliant — because these are not optional extras but the foundation of trustworthy science.

4

Career readiness

You will be prepared for roles in clinical genomics facilities, research, industry, and doctoral study — roles that require both deep technical skills and scientific communication.

Our Teaching Philosophy

Research-led



Teaching is grounded in active research. Many examples, datasets, and tools come from work happening in the labs of your lecturers — including data from PlantGenIE, the SwAsp population, and clinical genomics collaborations.

Hands-on from day one



You will do more exercises than you will listen to lectures. Every concept is introduced through doing. Exercises use real data — not toy examples — because that is where skills actually form.

Student-led development



The programme adapts to you. Your feedback shapes sessions in real time (via Mentimeter) and across the year (via course evaluations). If something is unclear, say so — we will address it.

Collaborative learning



You learn from each other as much as from your lecturers. Discussion, peer debugging, and shared repositories are all encouraged. Your cohort is one of your most valuable resources.

Progressive challenge



The programme is designed so that difficulty increases gradually. The first course assumes no prior command-line experience. By the thesis you will be conducting independent research.

Quality over coverage



We prefer depth over breadth. Better to genuinely understand ten concepts than to have encountered thirty. Where topics cannot be covered in the taught courses, we point to excellent self-study resources.

Programme Structure

Year 1

60 ECTS · 8 courses

- **Computing for Data-Driven Biology** 31 Aug–29 Sep 2026
- Data Science for Biology with R 30 Sep–1 Nov 2026
- Python Programming for Bioinformatics 2 Nov 2026–17 Jan 2027
- Statistical Methods in Bioinformatics 18 Jan–24 Mar 2027
- Functional Genomics Theory 25 Mar–2 May 2027
- Applied Functional Genomics 3 May–6 Jun 2027
- Comparative Genomics Year 1 · LP3–4
- Quantitative Biology and Integrative Omics Year 1 · LP3–4

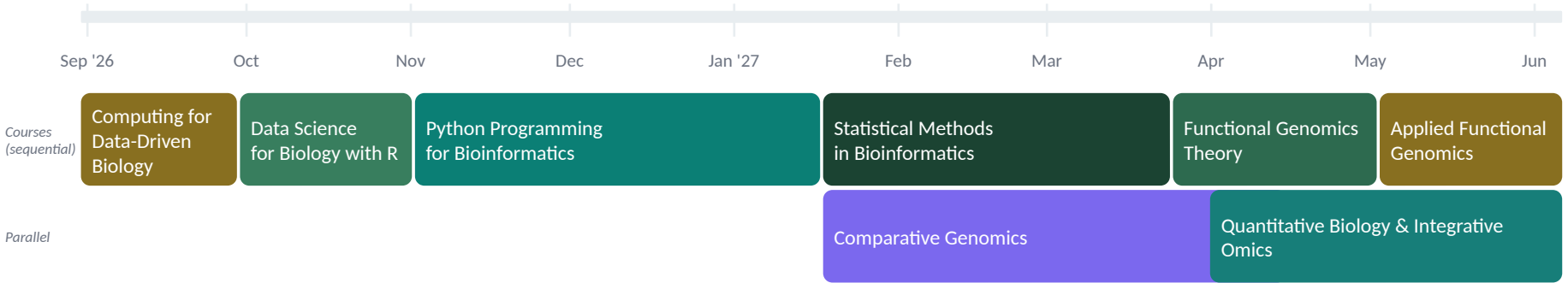
7.5 ECTS each (except thesis)

Year 2

60 ECTS · 4 courses + thesis

- Advanced Bioinformatics 30 Aug–29 Sep 2027
- Microbial Bioinformatics 30 Sep–1 Nov 2027
- Clinical Omics and Bioinformatics 2 Nov–2 Dec 2027
- Epigenetics and the Non-Coding Genome 3 Dec 2027–16 Jan 2028
- **Degree Project in Bioinformatics (thesis)** 30 ECTS · Spring 2028

Year 1 — Course Timeline



← You are here • 31 August 2026

Year 2 — Advanced Study and Research

7.5 hp

Advanced Bioinformatics

30 Aug – 29 Sep 2027

Deep-dives into algorithms, machine learning in bioinformatics, large-scale data integration.

7.5 hp

Microbial Bioinformatics

30 Sep – 1 Nov 2027

Metagenomics, amplicon sequencing, microbial ecology, clinical microbiology applications.

7.5 hp

Clinical Omics and Bioinformatics

2 Nov – 2 Dec 2027

Genomics in clinical settings: variant interpretation, cancer genomics, pharmacogenomics.

7.5 hp

Epigenetics and the Non-Coding Genome

3 Dec 2027 – 16 Jan 2028

DNA methylation, chromatin accessibility, regulatory non-coding RNAs, ATAC-seq, ChIP-seq.



Degree Project in Bioinformatics — 30 ECTS · Spring/Summer 2028

What You Will Be Able to Do



Computing

- ▶ Linux command line & HPC workflows
- ▶ Python and R for data analysis
- ▶ Version control with Git
- ▶ Container-based reproducible pipelines
- ▶ Slurm job scheduling



Genomics

- ▶ RNA-seq & differential expression
- ▶ Variant calling & interpretation
- ▶ Comparative & functional genomics
- ▶ Epigenomics (ChIP-seq, ATAC-seq)
- ▶ Single-cell sequencing



Data science

- ▶ Statistical modelling in bioinformatics
- ▶ Machine learning for omics data
- ▶ Dimensionality reduction & clustering
- ▶ Data visualisation
- ▶ Multi-omics integration



Applied domains

- ▶ Clinical genomics & variant reporting
- ▶ Metagenomics & microbiome analysis
- ▶ FAIR data management
- ▶ Bioinformatics service delivery
- ▶ Independent research (thesis)

Where This Takes You



Clinical Genomics

Bioinformatician at Clinical Genomics facilities (e.g. SciLifeLab, hospital genomics units). Variant interpretation, rare disease diagnosis, oncology sequencing.



PhD Research

Doctoral candidate in bioinformatics, genomics, systems biology, or a related field. Many of our graduates continue in academic research — in Sweden and internationally.



Industry

Pharmaceutical companies, biotech startups, sequencing companies (Illumina, Oxford Nanopore), data analytics, AI/ML companies working on biological data.



International Research Institutes

EMBL, Wellcome Sanger Institute, Broad Institute, Max Planck Institutes, and equivalents across Europe and globally. Our graduates work in all of these.

The programme is deliberately broad in its first year and specialist in its second — so you can develop an informed view of which direction suits you before committing.

Assessment, Grades, and Progression

The Swedish grade scale



Underkänd (Fail)

Does not meet the learning outcomes. May resubmit once.



Godkänd (Pass)

Meets all learning outcomes at the expected level.



Väl Godkänd (Pass with Distinction)

Exceeds expectations with demonstrable critical analysis and independent judgement.

Progression requirements

- ▶ Pass all Year 1 courses before starting Year 2
- ▶ Thesis registration requires at least 45 ECTS passed from Year 2 taught courses
- ▶ Resubmission is available for any failed assessment — typically once, with a deadline set by the examiner
- ▶ Grade of G or VG on each course — no weighted average in Swedish university grading
- ▶ If you are struggling: contact Nathaniel early. We have processes to support you.



AI Tools Policy

You may use AI coding and writing tools. You must understand any AI-generated code before running it. Assessed work must include an AI use declaration if AI was involved. Critical thinking and independent analysis cannot be delegated.

Your First Week — What to Expect

Computing for Data-Driven Biology begins 31 August 2026.

Week 1 **31 Aug – 3 Sep**

Linux, HPC Environment + What is Bioinformatics

HPC2N-led sessions on Linux, the command line, and the Kebnekaise environment, interleaved with the course leader's opening session on what bioinformatics is.

Week 2 **8 – 11 Sep**

Git, Software + FAIR and Open Science

HPC2N sessions on Git and building/accessing software on Kebnekaise, followed by the course leader's FAIR and Open Science lecture and its hands-on PlantGenIE exercises.

Week 3 **15 – 18 Sep**

Databases, File Formats + Batch/Slurm

Biological databases and file formats (FASTQ, BAM, VCF, GTF), then HPC2N's introduction to the Slurm batch system and a summary exercises/catch-up session.

Week 4 **21 – 29 Sep**

Independent Work + Exam

No scheduled sessions until the exam. Use this time for the FAIR essay (deadline 29 September, 17:00). Exam: 29 September, 13:00-15:00.

Before you arrive: log in to Kebnekaise via OnDemand (instructions emailed). No other preparation is required.

Questions?

If you think of something later:



nathaniel.street@umu.se

Course and programme questions



Canvas discussion board

Questions others may also have



KBC Building office

Drop in — door is usually open

Welcome to the programme. We are glad you are here.